

“Heaven’s Light is Our Guide”

Rajshahi University of Engineering & Technology

Department of **Electrical and Computer Engineering**



Lab Report

Course Code : ECE 2112

Course Name : Digital Techniques Sessional

Lab No : 02

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Objective:

To design and simulate a logic circuit from a given Boolean expression using Logisim Evolution and verify its output with the corresponding truth table.

Theory:

A Boolean expression is a mathematical representation of a digital logic function using binary variables and logical operators such as AND ($\cdot$), OR ($+$), and NOT ($'$). These expressions can be implemented using combinations of basic logic gates to create digital circuits. By simplifying Boolean expressions, the required logic circuit can be made more efficient with fewer gates. Simulation software like Logisim Evolution allows users to construct the circuit, test different input combinations, and verify that the output matches the expected truth table. This process helps in understanding the relationship between Boolean algebra and digital logic circuit design.

Experiment 1:

$$F = AB + \bar{B}C + AC + \bar{A}B\bar{D}$$

Boolean Expression Simplification:

$$F = AB + \bar{B}C + AC + \bar{A}B\bar{D}$$

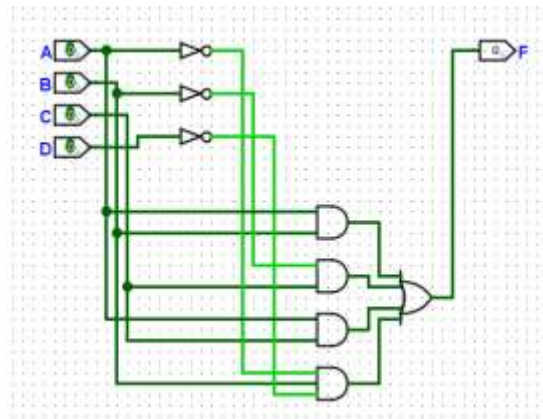
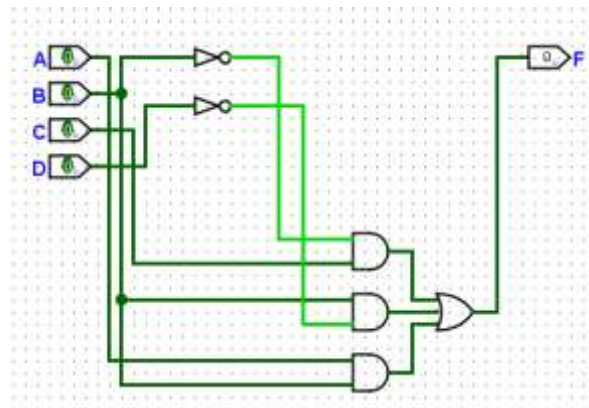
$$= B(A + \bar{A}\bar{D}) + \bar{B}C + AC$$

$$= B(A + \bar{D}) + \bar{B}C + AC$$

$$= AB + B\bar{D} + \bar{B}C + AC$$

$$= AB + \bar{B}C + B\bar{D}$$

$$\therefore F = AB + \bar{B}C + B\bar{D}$$

Circuit Diagram:Figure 1: $F = AB + \bar{B}C + AC + \bar{A}B\bar{D}$ Figure 2: $F = AB + \bar{B}C + B\bar{D}$ **Truth Table:**

$$F = AB + \bar{B}C + AC + \bar{A}B\bar{D}$$

A	B	C	D	F
0	0	0	0	0
0	0	0	1	0
0	0	1	0	1
0	0	1	1	1
0	1	0	0	1
0	1	0	1	0
0	1	1	0	1

$$F = AB + \bar{B}C + B\bar{D}$$

A	B	C	D	F
0	0	0	0	0
0	0	0	1	0
0	0	1	0	1
0	0	1	1	1
0	1	0	0	1
0	1	0	1	0
0	1	1	0	1

0	1	1	1	0
1	0	0	0	0
1	0	0	1	0
1	0	1	0	1
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

0	1	1	1	0
1	0	0	0	0
1	0	0	1	0
1	0	1	0	1
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

Experiment 2:

$$F = [(\bar{A} + B)(C + \bar{D})] \cdot [A\bar{B} + C + \bar{D}]$$

Boolean Expression Simplification:

$$F = [(\bar{A} + B)(C + \bar{D})] \cdot [A\bar{B} + C + \bar{D}]$$

$$[(\bar{A} + B)(C + \bar{D})] = (\bar{A} + B) + (C + \bar{D}) = A\bar{B} + \bar{C}D$$

$$= (A\bar{B} + \bar{C}D)(A\bar{B} + C + \bar{D})$$

$$= A\bar{B}(A\bar{B} + C + \bar{D}) + \bar{C}D(A\bar{B} + C + \bar{D})$$

$$= A\bar{B} + A\bar{B}C + A\bar{B}\bar{D} + A\bar{B}\bar{C}D + \bar{C}CD + \bar{C}D\bar{D}$$

$$= A\bar{B} + A\bar{B}C + A\bar{B}\bar{D} + A\bar{B}\bar{C}D + 0 + 0$$

$$= A\bar{B}(1 + C + \bar{D}) + A\bar{B}\bar{C}D$$

$$= A\bar{B} + A\bar{B}\bar{C}D$$

$$= A\bar{B}(1 + \bar{C}D)$$

$$= A\bar{B}$$

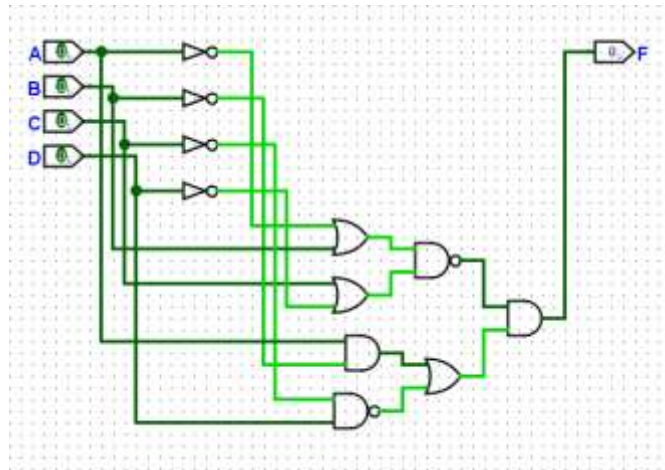
Circuit Diagram:

Figure 3: $F = [(\bar{A} + B)(C + \bar{D})] \cdot [\bar{A}B + C + \bar{D}]$

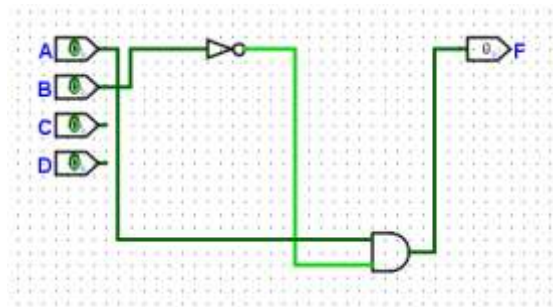


Figure 4: $F = \bar{A}\bar{B}$

Truth Table:

$$F = [(\bar{A} + B)(C + \bar{D})] \cdot [\bar{A}B + C + \bar{D}]$$

A	B	C	D	F
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0

$$F = \bar{A}\bar{B}$$

A	B	F
0	0	0
0	1	0
1	0	1
1	1	0

0	1	1	0	0
0	1	1	1	0
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0

Experiment 3:

$$F = ABCD + \bar{A}BCD + A\bar{B}CD + AB\bar{C}D + ABC\bar{D}$$

Boolean Expression Simplification:

$$F = ABCD + AB\bar{C}D + \bar{A}BCD + A\bar{B}CD + ABC\bar{D}$$

$$= ABD(C + \bar{C}) + \bar{A}BCD + A\bar{B}CD + ABC\bar{D}$$

$$= ABD + \bar{A}BCD + A\bar{B}CD + ABC\bar{D}$$

$$= ABD + CD(\bar{A}B + A\bar{B}) + ABC\bar{D}$$

$$= AB(D + C\bar{D}) + CD(\bar{A}B + A\bar{B})$$

$$= AB(D + C) + CD(A \oplus B)$$

$$= ABD + ABC + CD(A \oplus B)$$

$$\therefore F = ABD + ABC + CD(A \oplus B)$$

Circuit Diagram:

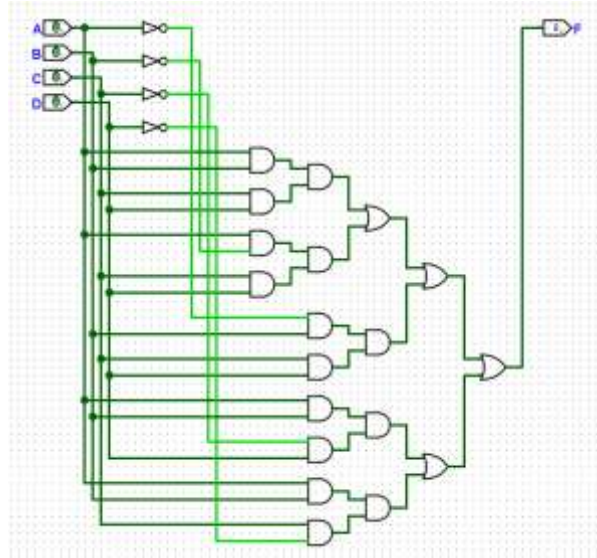


Figure 5: $F = ABCD + AB\bar{C}D + \bar{A}BCD + A\bar{B}CD + ABC\bar{D}$

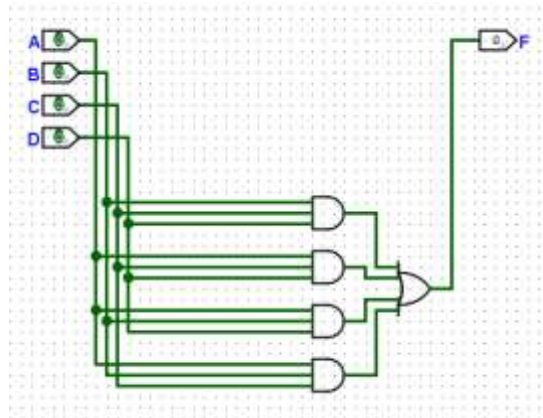


Figure 6: $F = ABD + ABC + CD(A \oplus B)$

Truth Table:

$$F = ABCD + AB\bar{C}D + \bar{A}BCD + A\bar{B}CD + ABC\bar{D}$$

A	B	C	D	F
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0

$$F = ABD + ABC + CD(A \oplus B)$$

A	B	C	D	F
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0

0	1	0	1	0
0	1	1	0	0
0	1	1	1	1
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	1	0	0	0
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

0	1	0	1	0
0	1	1	0	0
0	1	1	1	1
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	1	0	0	0
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

Experiment 4:

$$F = (A + B + C + D)(A + B + \bar{C})(A + \bar{B} + C)(\bar{A} + B + C)$$

Boolean Expression Simplification:

$$\begin{aligned}
 F &= (A + B + C + D)(A + B + \bar{C})(A + \bar{B} + C)(\bar{A} + B + C) \\
 &= [(A + B) + (C + D)\bar{C}] [C + (A + \bar{B})(\bar{A} + B)] \\
 &= [A + B + C\bar{C} + \bar{C}D] [C + (A\bar{A} + AB + \bar{A}\bar{B} + B\bar{B})] \\
 &= (A + B + \bar{C}D) (C + AB + \bar{A}\bar{B}) \\
 &= A(C + AB + \bar{A}\bar{B}) + B(C + AB + \bar{A}\bar{B}) + \bar{C}D(C + AB + \bar{A}\bar{B}) \\
 &= (AC + AAB + A\bar{A}\bar{B}) + (BC + BAB + \bar{A}B\bar{B}) + (C\bar{C}D + AB\bar{C}D + \bar{A}\bar{B}\bar{C}D) \\
 &= AC + AB + BC + AB + AB\bar{C}D + \bar{A}\bar{B}\bar{C}D \\
 &= AB + AC + BC + AB\bar{C}D + \bar{A}\bar{B}\bar{C}D
 \end{aligned}$$

$$= AB(1 + \bar{C}D) + AC + BC + \bar{A}\bar{B}\bar{C}D$$

$$= AB + AC + BC + \bar{A}\bar{B}\bar{C}D$$

$$\therefore F = AB + AC + BC + \bar{A}\bar{B}\bar{C}D$$

Circuit Diagram:

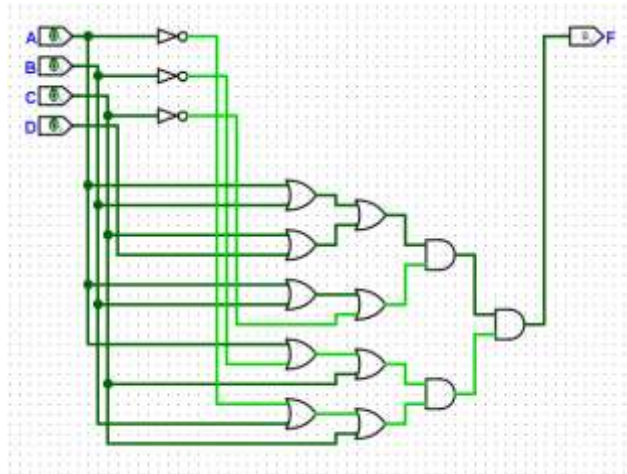


Figure 7: $F = (A + B + C + D)(A + B + \bar{C})(A + \bar{B} + C)(\bar{A} + B + C)$

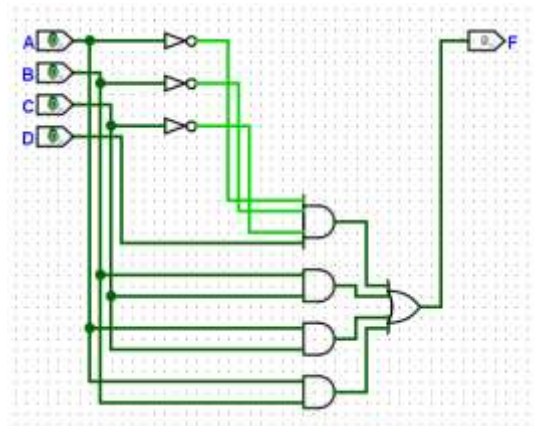


Figure 8: $F = AB + AC + BC + \bar{A}\bar{B}\bar{C}D$

Truth Table:

$$F = (A + B + C + D)(A + B + \bar{C})(A + \bar{B} + C)(\bar{A} + B + C)$$

A	B	C	D	F
0	0	0	0	0
0	0	0	1	1
0	0	1	0	0

$$F = AB + AC + BC + \bar{A}\bar{B}\bar{C}D$$

A	B	C	D	F
0	0	0	0	0
0	0	0	1	1
0	0	1	0	0

0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	1
0	1	1	1	1
1	0	0	0	0
1	0	0	1	0
1	0	1	0	1
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	1
0	1	1	1	1
1	0	0	0	0
1	0	0	1	0
1	0	1	0	1
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

Experiment 5:

$$F = [A + \bar{B}(C + D)] [\bar{A} + B(\bar{C} + \bar{D})]$$

Boolean Expression Simplification:

$$F = [A + \bar{B}(C + D)] [\bar{A} + B(\bar{C} + \bar{D})]$$

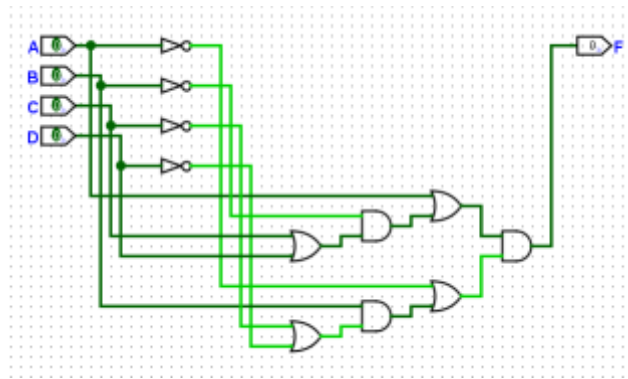
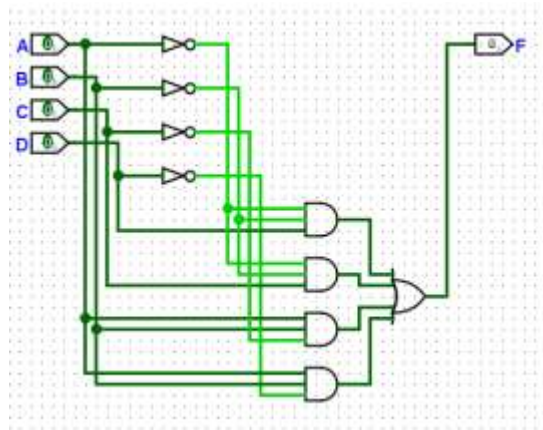
$$= A\bar{A} + AB(\bar{C} + \bar{D}) + \bar{A}\bar{B}(C + D) + B\bar{B}(C + D)(\bar{C} + \bar{D})$$

$$= 0 + AB(\bar{C} + \bar{D}) + \bar{A}\bar{B}(C + D) + 0$$

$$= AB(\bar{C} + \bar{D}) + \bar{A}\bar{B}(C + D)$$

$$= AB\bar{C} + AB\bar{D} + \bar{A}\bar{B}C + \bar{A}\bar{B}D$$

$$\therefore F = AB\bar{C} + AB\bar{D} + \bar{A}\bar{B}C + \bar{A}\bar{B}D$$

Circuit Diagram:Figure 9: $[A + \bar{B}(C + D)][\bar{A} + B(\bar{C} + \bar{D})]$ Figure 10: $F = AB\bar{C} + AB\bar{D} + \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}\bar{D}$ **Truth Table:**

$$[A + \bar{B}(C + D)][\bar{A} + B(\bar{C} + \bar{D})]$$

A	B	C	D	F
0	0	0	0	0
0	0	0	1	1
0	0	1	0	1
0	0	1	1	1
0	1	0	0	0

$$F = AB\bar{C} + AB\bar{D} + \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}\bar{D}$$

A	B	C	D	F
0	0	0	0	0
0	0	0	1	1
0	0	1	0	1
0	0	1	1	1
0	1	0	0	0

0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	0
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	0

0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	0
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	0

Discussion:

The experiment successfully demonstrated the implementation of a logic circuit from a given Boolean expression using Logisim Evolution. The designed circuit accurately produced the expected outputs for all possible input combinations, confirming the correctness of both the Boolean expression and the circuit design. The simulation results matched the theoretical truth table, verifying the proper operation of the implemented logic gates. This experiment highlights the importance of Boolean algebra in designing digital systems and shows how logic circuit simulation can be used to analyze, verify, and troubleshoot digital circuits before hardware implementation.

Handwritten Simplifications:

